



BATTERED SHORE

The highest waves (from trough to crest) are produced by high winds blowing across the ocean for long distances uninterrupted by land. For example, the southwest coast of England gets some of the biggest storm waves because they come right across the Atlantic Ocean. Waves crashing against the shore weaken cliffs, and cause some to fall into the sea.

Vane to orient buoy into the wind

Flashing light can be seen by ships

Antenna for transmitting satellite signals

Propeller wind vane measures speed and direction of wind

Sensor measures air temperature

Tower measures 10 ft (3 m) high

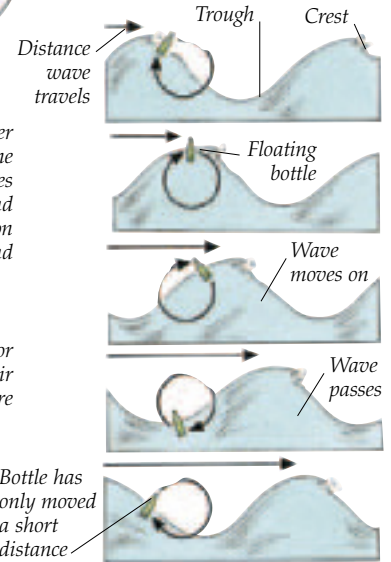
Sensor measures temperature of sea's surface

Hull of buoy is 10 ft (3 m) across, so is big enough for a person to board it

Buoy for monitoring winds and temperatures of air and sea surface

MONITORING BUOY

This buoy is moored to the seabed and floats on the ocean's surface. Its instruments measure the winds and temperatures of the air and sea surface. Readings collected by these instruments are stored on board the buoy in computer systems. Its data is sent via satellite to a land-based station. The buoy's location can be found by the signals it sends back to the satellite. Close to, its radio signals and flashing lights can be identified by ships.



MAKING WAVES

Waves are formed due to the action of wind on the surface of the water, which causes friction.

While the wave itself moves forward, pushed in the same direction as the wind, the water making up the wave hardly shifts at all. It follows a circular path, shown by the floating bottle (above). When the wind is stronger, the waves may spill over at the top and break into frothy spray. Waves, driven by winds toward a beach, break when the water becomes too shallow.